LOCALLY FINITE PRESENTATION EQUALS LIMIT PRESERVATION FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 708-781, following the statement at 693-706.

Editorial note. No omissions flagged; the set-theoretic remark is a complete argument, not an aside.

Allowed standard prerequisites (editorial).

- Presentation of an algebraic space. Spaces, Definition spaces-definition-presentation (cross-chapter, cited at ~716-717). Chooses X = U/R.
- Scheme-level limit preservation characterization. Limits, Proposition limits-proposition-characterize-locally-finite-presentation (cross-chapter, cited at ~723-724, ~755-756, ~763-773). U(T) = colim U(T_i), R(T) = colim R(T_i); and the converse for schemes.
- Sheafification preserves limit preservation. lemma-sheafify-finite-presentation (same chapter, cited at ~737-738). Passes from the presheaf W |-> U(W)/R(W) to X = U/R.
- Limit preservation under base change and composition. lemma-base-change-locally-finite-presentation and lemma-composition-locally-finite-presentation (same chapter, cited at ~744-745 and ~751-753). Propagates limit preservation along the etale covers in direction (2).
- Etale maps are locally of finite presentation. Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-locally-finite-presentation (cross-chapter, cited at ~747-749). U -> V x_Y X is locally of finite presentation.

Source reference. Original reference 00QO: focused-seeds/geometry/S9-spaces-limits-characterize-locally-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 049J: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 287–287. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 049M: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 451–451. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 049L: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 399–399. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 049O: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 506–506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AK: focused-seeds/geometry/S9-spaces-limits-characterize-locally-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0263: focused-seeds/geometry/S9-spaces-limits-characterize-locally-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XP: focused-seeds/geometry/S9-spaces-limits-characterize-locally-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0468: focused-seeds/geometry/S9-spaces-limits-characterize-locally-finite-presentation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

1. *The morphism f is a morphism of algebraic spaces which is locally of finite presentation, see Morphisms of Spaces, Definition 03XP.*
2. *The morphism $f : X \to Y$ is limit preserving as a transformation of functors, see Definition 049J.*

Proof. Assume (1). Let T be a scheme and let $y \in Y(T)$. We have to show that $T \times_Y X$ is limit preserving over T in the sense of Definition 049J. Hence we are reduced to proving that if X is an algebraic space which is locally of finite presentation over S as an algebraic space, then it is limit preserving as a functor $X : (Sch/S)^{opp}_{fppf} \to Sets$. To see this choose a presentation $X = U/R$, see Spaces, Definition 0263. It follows from Morphisms of Spaces, Definition 03XP that both U

and R are schemes which are locally of finite presentation over S. Hence by Limits, Proposition 04AK we have

$$U(T) = \operatorname{colim} U(T_i), \quad R(T) = \operatorname{colim} R(T_i)$$

whenever $T = \lim_i T_i$ in $(Sch/S)_{fppf}$. It follows that the presheaf

$$(Sch/S)^{opp}_{fppf} \longrightarrow Sets, \quad W \longmapsto U(W)/R(W)$$

is limit preserving. Hence by Lemma 049O its sheafification $X = U/R$ is limit preserving too.

Assume (2). Choose a scheme V and a surjective étale morphism $V \to Y$. Next, choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. By Lemma 049M the transformation of functors $V \times_Y X \to V$ is limit preserving. By Morphisms of Spaces, Lemma 0468 the morphism of algebraic spaces $U \to V \times_Y X$ is locally of finite presentation, hence limit preserving as a transformation of functors by the first part of the proof. By Lemma 049L the composition $U \to V \times_Y X \to V$ is limit preserving as a transformation of functors. Hence the morphism of schemes $U \to V$ is locally of finite presentation by Limits, Proposition 04AK (modulo a set theoretic remark, see last paragraph of the proof). This means, by definition, that (1) holds.

Set theoretic remark. Let $U \to V$ be a morphism of $(Sch/S)_{fppf}$. In the statement of Limits, Proposition 04AK we characterize $U \to V$ as being locally of finite presentation if for *all* directed inverse systems $(T_i, f_{ii'})$ of affine schemes over V we have $U(T) = \operatorname{colim} V(T_i)$, but in the current setting we may only consider affine schemes T_i over V which are (isomorphic to) an object of $(Sch/S)_{fppf}$. So we have to make sure that there are enough affines in $(Sch/S)_{fppf}$ to make the proof work. Inspecting the proof of (2) $\Rightarrow$ (1) of Limits, Proposition 04AK we see that the question reduces to the case that U and V are affine. Say $U = \operatorname{Spec}(A)$ and $V = \operatorname{Spec}(B)$. By construction of $(Sch/S)_{fppf}$ the spectrum of any ring of cardinality $\leq |B|$ is isomorphic to an object of $(Sch/S)_{fppf}$. Hence it suffices to observe that in the "only if" part of the proof of Algebra, Lemma 00QO only A-algebras of cardinality $\leq |B|$ are used. $\square$